Lian Cataytay

Data analyst and economist · Dashboards, models and economic research

BUSINESS & INTELLIGENCE · MACHINE LEARNING & RISK · ECONOMICS & RESEARCH

A data analyst who asks an economist's questions

Twelve projects in business intelligence, machine learning and economic research, each with the data and charts behind it.

I'm a data analyst and economist from Davao City in the Philippines. My work covers econometrics, business intelligence and machine learning, and my projects include retention analytics, churn prediction, credit risk and workforce analysis.

I work in Python, SQL, R, Stata, Power BI, Tableau and a lot of Excel, taking raw data through to dashboards and models that people can follow and use.

CONTENTS

Lead project

CARP and Agricultural Productivity: A Synthetic Control Study 3

Business & Intelligence

E-commerce Customer Retention Analytics 3

eCommerce Marketplace Ads Performance Dashboard 4

FP&A Revenue, Cost & Expense Forecasting Dashboard 4

CRM & Event Marketing Data Hygiene System 5

Employee Attrition Risk and Workforce Intelligence Analysis 5

Machine Learning & Risk

AI Customer Churn Intelligence System 6

Banking Risk and Financial Performance Analysis 6

Customer Sentiment Intelligence Dashboard 7

Economics & Research

Philippine Emigration ARDL Macroeconomic Analysis 7

Philippines Labor Market Trend Analysis 8

Media Insight Report: Philippine Inflation and Policy Debate 8

BY THE NUMBERS

3+
years in analytics and research

12
projects from data to findings

11
with public code on GitHub

1.30
CGPA, BS Economics, University of Southeastern Philippines

TOOLKIT

Python · SQL · R · Stata · Excel · Power BI · Tableau · Pandas · Scikit-learn · SHAP

CONTACT

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Curriculum vitae

EXPERIENCE

Aug 2025 – Oct 2025

Data Analyst Intern
Socio-Economic Research and Data Analytics Center

Standardized demographic data from several sources in RStudio and Excel, built charts and Excel dashboards, and wrote up the statistics as briefs for policymakers and program staff.

Oct 2024 – Present

Freelance Data Analyst
Self-employed

Revenue and expense trackers with budget-versus-actual checks, KPI dashboards for 8+ marketing and finance metrics, and regression-based scenario and A/B lift models.

Aug 2024 – Jan 2026

Econometrics and Applied Economics Research Analyst
University of Southeastern Philippines

Built Prais-Winsten, time-series, logit and probit models for research on the macroeconomy, inflation and loan delinquency, and wrote up the findings for policy readers.

Aug 2022 – Aug 2024

Lead Feature Editor
The Collegiate Headlight, USeP

Wrote, edited and checked 100+ published pieces, and coached junior writers on using evidence and structuring a story.

Aug 2020 – Apr 2022

Administrative Assistant
Saint John Marie Vianney Academy of Sulop, Inc.

Kept accounting logs, records and reports, and made 50+ educational and promotional materials.

EDUCATION

University of Southeastern Philippines

BS Economics, major in Development Economics
Davao City · CGPA 1.30

Coursework in econometrics, time-series analysis, financial economics, development economics, statistics, and project evaluation.

TOOLKIT

Python · SQL · R · Stata · Excel · Power BI · Tableau · Pandas · Scikit-learn · SHAP

LEAD PROJECT · AGRICULTURAL ECONOMICS · SYNTHETIC CONTROL METHOD · IMPACT EVALUATION

CARP and Agricultural Productivity: A Synthetic Control Study

Did the 1988 agrarian reform change farm productivity? A synthetic control comparison for labor, land and TFP.

Context. The Comprehensive Agrarian Reform Program (CARP) began in 1988 and still shapes Philippine debates on equity and rural development. Its effect on national farm productivity has rarely been tested against a credible counterfactual.

Method. I used the synthetic control method to build a "synthetic Philippines" that tracks the real one before 1988, then compared the two on labor productivity, land productivity and TFP after the reform. A placebo run on GDP per capita checked whether the gaps were specific to agriculture.

Findings. From 1989 to 2008 Philippine agriculture ran below its synthetic control: labor productivity by about 18.2%, land productivity by 16.1% and TFP by about 9%. GDP per capita was only 2.5% below, so the shortfall belongs to agriculture, and most of it comes from slower TFP growth.

Tools: R, R Markdown, Synthetic control, TFP analysis
Code: github.com/liancataytayjk-coder/agricultural-scm-productivity-analysis

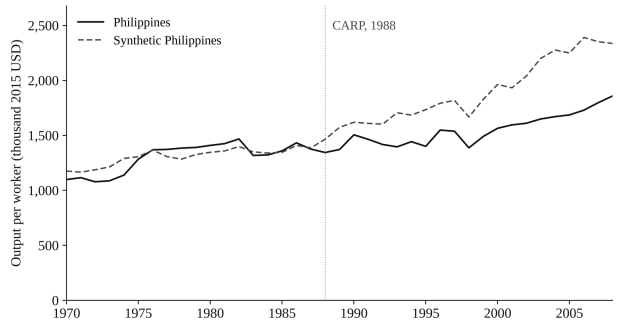


Figure 1. Labor productivity: the Philippines and its synthetic control, 1970 to 2008. *After 1988 the Philippines averages about 18.2% lower than its synthetic control (1989 to 2008). Source: Cataytay, Deluna and Evangelio, synthetic control analysis.*

BUSINESS & INTELLIGENCE · CUSTOMER ANALYTICS · RETENTION · SEGMENTATION

E-commerce Customer Retention Analytics

Cohorts and customer segments from the Olist marketplace, built to show where retention effort pays off.

Context. I wanted to know which customers bring in the most value and which are most likely to leave.

Method. Cohort analysis and segmentation in Python and Pandas, then dashboards in Tableau.

Findings. Across 93,358 customers and $19.9M in revenue, 59.1% churned. Champions were only 6.9% of customers and 14.0% were at risk, which tells you where retention effort should start.

Tools: Python, Pandas, Tableau, Cohort analysis
Code: github.com/liancataytayjk-coder/ecommerce-customer-retention-analytics

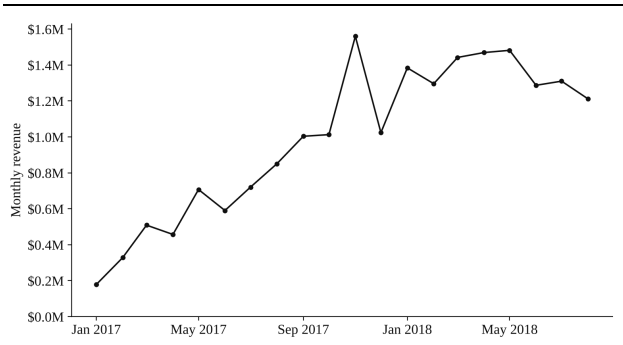


Figure 2. Monthly marketplace revenue, 2017 to 2018. *Months with fewer than 100 orders are omitted. Olist Brazilian e-commerce data. Source: E-commerce Customer Retention Analytics repository.*

BUSINESS & INTELLIGENCE · MARKETPLACE ADS · ACOS/TACOS · BUDGET ALLOCATION

eCommerce Marketplace Ads Performance Dashboard

Ads data from Amazon, Google, TikTok and eBay in one model, with ACoS/TACoS, stock checks and budget calls.

Context. Marketplace ad teams often juggle separate exports from Amazon Ads, Google Ads, TikTok Ads and eBay Ads, which makes it hard to compare spend, revenue, profit and stock across them.

Method. I merged the four platforms into one schema, cleaned SKU and campaign fields, joined product and inventory data, calculated ACoS/TACoS-style KPIs and added checks that catch reporting errors.

Findings. The model covers 42,360 ad rows and 6,500 sales rows, with $86.7M in attributed ad revenue and an average ROAS of 6.80. It flagged 4,903 campaign and SKU rows to cut or pause.

Tools: Excel, SQL, Python, Streamlit
Code: github.com/liancataytayjk-coder/ecommerce-marketplace-ads-performance-dashboard

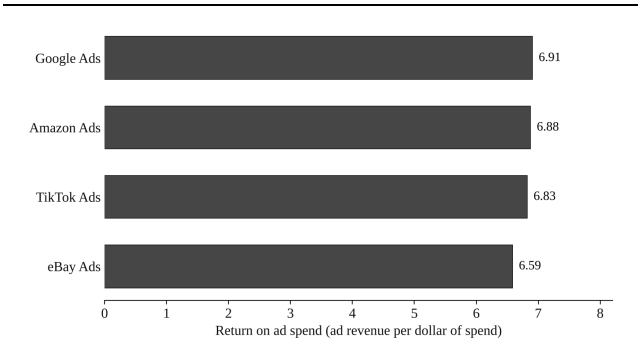


Figure 3. Return on ad spend by marketplace platform. *Simulated ad exports across four platforms. Source: eCommerce Marketplace Ads Performance Dashboard repository.*

BUSINESS & INTELLIGENCE · FP&A · FORECASTING · VARIANCE ANALYSIS

FP&A Revenue, Cost & Expense Forecasting Dashboard

Monthly budget-versus-actual reporting, forecasts, scenarios and an anomaly list for month-end review.

Context. Finance teams need monthly reporting they can trust on revenue, costs, expenses, forecast accuracy and KPIs.

Method. I cleaned the finance data, calculated variances against budget and against forecast, built scenario forecasts, tracked KPIs and flagged anomalies.

Findings. Over 36 months the model covers $281.6M in revenue at a 50.8% gross margin, with 98.2% forecast accuracy. It found 71 unfavorable variances and 369 anomalies for review.

Tools: Excel, SQL, Python, Streamlit
Code: github.com/liancataytayjk-coder/fpa-revenue-cost-expense-forecasting-dashboard

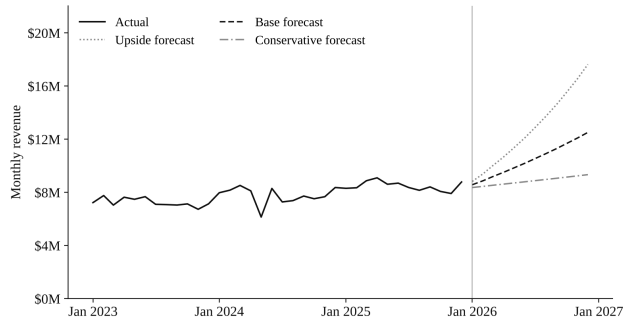


Figure 4. Monthly revenue, 2023 to 2025, with 2026 forecast scenarios. *Synthetic financials. Source: FP&A Revenue, Cost & Expense Forecasting Dashboard repository.*

BUSINESS & INTELLIGENCE · CRM HYGIENE · EVENT MARKETING · SALES OPERATIONS

CRM & Event Marketing Data Hygiene System

Cleans a messy CRM export: duplicates merged, contacts checked, and event follow-ups ready to send.

Context. Marketing and sales teams often work from CRM exports full of duplicate contacts, bad emails, inconsistent phone numbers and half-filled event records.

Method. I standardized contact fields, scored how complete each record was, merged duplicate groups, matched event registrations to contacts and prepared the follow-up lists.

Findings. 1,500 raw records became 1,187 unique contacts after 313 duplicates were removed. 469 emails were flagged as invalid, and 310 referrals were ready for follow-up.

Tools: Excel, Power Query logic, Python, Pandas
Code: github.com/liancataytayjk-coder/crm-event-marketing-data-hygiene-system

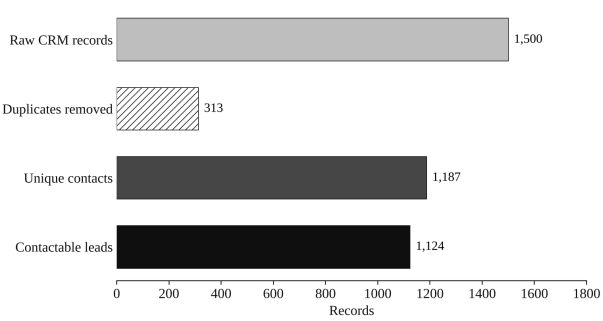


Figure 5. From raw export to contactable leads. *469 invalid emails and 283 invalid phone numbers were also flagged. Source: CRM & Event Marketing Data Hygiene System repository.*

BUSINESS & INTELLIGENCE · HR ANALYTICS · ATTRITION RISK · WORKFORCE INTELLIGENCE

Employee Attrition Risk and Workforce Intelligence Analysis

Which teams are most likely to lose people, scored from HR and performance data.

Context. I wanted to see whether workforce data could point to the employees and teams most likely to leave.

Method. Attrition risk scoring and hotspot analysis in Python and Pandas.

Findings. Estimated attrition risk was 21.6% across the employee groups analyzed, and 9 teams stood out enough to flag for manager review.

Tools: Python, Pandas, Risk scoring, Hotspot analysis
Code: not public

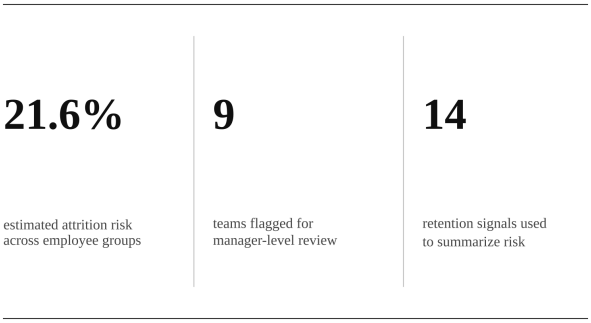


Figure 6. Key figures from the workforce attrition analysis. *Summary figures as reported in the project, which has no public repository.*

MACHINE LEARNING & RISK · MACHINE LEARNING · EXPLAINABILITY · CHURN RISK

AI Customer Churn Intelligence System

A churn model that explains itself: risk scores, SHAP drivers and a Power BI view for the retention team.

Context. I wanted a churn model that is accurate and can also say why it flagged each customer.

Method. Classification models in scikit-learn, SHAP to explain each prediction, and Power BI for the reporting.

Findings. The model scored 1,409 customers. The ones it placed in the High Risk group churned at 78.8%; the Low Risk group churned at 12.9%. Contract type, pricing and service usage drove most of the risk, and $27,972 in revenue was flagged as at risk.

Tools: Python, scikit-learn, SHAP, Power BI
Code: github.com/liancataytayjk-coder/ai-customer-churn-intelligence-system

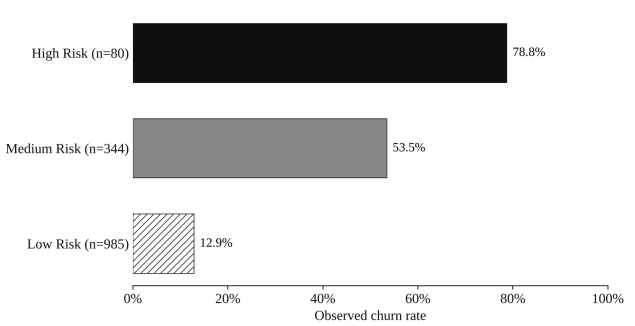


Figure 7. Observed churn by model risk segment. *1,409 customers scored by the model. Source: AI Customer Churn Intelligence System repository.*

MACHINE LEARNING & RISK · BANKING ANALYTICS · CREDIT RISK · FINANCIAL PERFORMANCE

Banking Risk and Financial Performance Analysis

Which borrowers default? A classification model and five risk bands on a 2,000-loan portfolio.

Context. I wanted to find which borrower and loan traits go with default.

Method. A classification model in Python and Pandas, with feature importance. I then split the loans into five equal risk bands and estimated the expected loss in each.

Findings. The model reached a ROC AUC of 0.72. Loans in the Very High band defaulted at 53% and those in the Very Low band at 2%, and the Very High band held 53% of expected-loss exposure. Credit score and interest rate mattered most.

Tools: Python, Pandas, Classification analytics, Feature importance
Code: github.com/liancataytayjk-coder/banking-risk-and-financial-performance-analysis

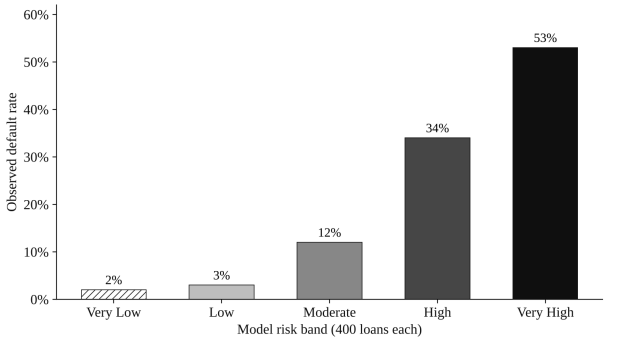


Figure 8. Observed default rate by model risk band. *Synthetic 2,000-loan portfolio; overall default rate 20.8%. Source: Banking Risk and Financial Performance Analysis repository.*

MACHINE LEARNING & RISK · NLP · SENTIMENT ANALYSIS · BUSINESS INTELLIGENCE

Customer Sentiment Intelligence Dashboard

Turns 500 customer reviews in more than one language into sentiment, themes and an NPS a manager can act on.

Context. I wanted to see how far raw customer reviews could go toward something a manager would actually use.

Method. Cleaning and sentiment analysis in Python, then KPIs prepared for Power BI.

Findings. 41.6% of the reviews were positive, 33.0% neutral and 25.4% negative, for an NPS of 30. Coffee quality, staff behavior and amenities came up most often, and high-risk reviews were flagged for follow-up.

Tools: Python, Power BI, Pandas, NLP workflow
Code: github.com/liancataytayjk-coder/customer-sentiment-intelligence-dashboard

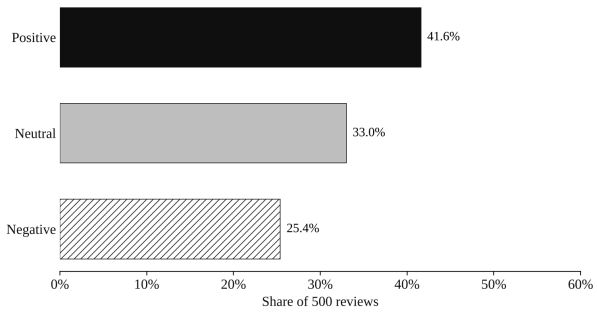


Figure 9. Sentiment of 500 multilingual customer reviews. *Net Promoter Score of 30. Source: Customer Sentiment Intelligence Dashboard repository.*

ECONOMICS & RESEARCH · MACROECONOMICS · MIGRATION · ARDL MODELING

Philippine Emigration ARDL Macroeconomic Analysis

ARDL models of how income, government spending, labor share and the exchange rate relate to Philippine emigration.

Context. I wanted to see how the Philippine economy relates to emigration over time.

Method. ARDL bounds tests, error-correction models and time-series diagnostics in Stata.

Findings. Five of seven ARDL specifications found a long-run relationship. In the baseline model, a 1% rise in Philippine GDP per capita goes with a 0.46% fall in emigration in the long run. Error-correction terms of −0.56 to −1.00 mean most shocks fade within a year. COVID-19 shows up as a large downward shift.

Tools: Stata, ARDL bounds testing, ECM modeling, Time-series diagnostics
Code: github.com/liancataytayjk-coder/philippine-emigration-ardl-macroeconomic-analysis

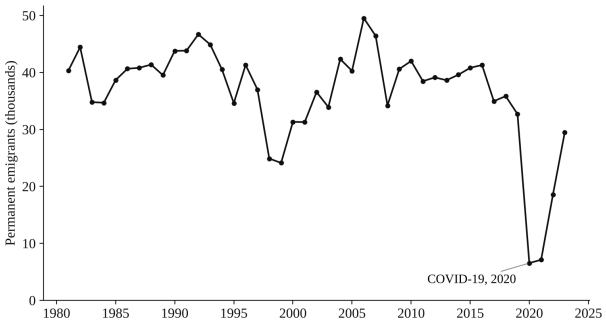


Figure 10. Permanent Philippine emigrants, 1981 to 2023. *Permanent emigrant counts. Source: project repository.*

ECONOMICS & RESEARCH · LABOR ECONOMICS · TIME-SERIES TRENDS · POLICY INSIGHT

Philippines Labor Market Trend Analysis

Employment, underemployment and the shift between sectors in Philippine labor force data.

Context. I wanted to see how employment in the Philippines has changed, and how good the jobs are.

Method. Trend analysis of the series in Python and Pandas.

Findings. In 2025 labor force participation was 64.1%, unemployment 4.2% and underemployment 11.4%. Underemployment stays at almost three times unemployment, so job quality is a bigger gap than job availability.

Tools: Python, Pandas, Time-series analysis, Excel
Code: github.com/liancataytayjk-coder/philippines-labor-market-trend-analysis

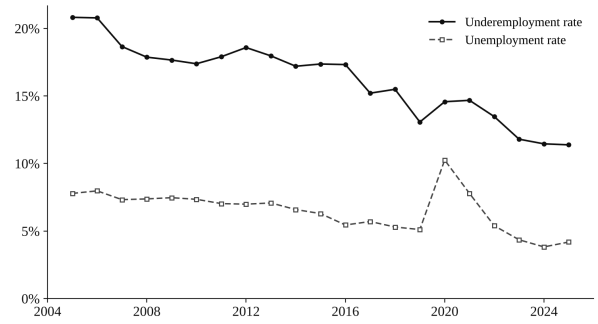


Figure 11. Unemployment and underemployment rates, 2005 to 2025. *Annual figures from Labor Force Survey indicators. Source: project repository.*

ECONOMICS & RESEARCH · MEDIA INSIGHTS · THEMATIC ANALYSIS · POLICY RISK

Media Insight Report: Philippine Inflation and Policy Debate

How news coverage of the 2026 inflation spike turns into risks and openings for a retail client.

Context. Philippine headline inflation rose from 2.0% in January to 7.2% in April 2026, its highest in three years. I wanted to show how news coverage of that rise turns into specific risks and openings for a business.

Method. I coded the coverage into three themes (cost-of-living hardship, wage justice against business costs, and social protection as a shock absorber), read its tone, and mapped stakeholders for a retail and FMCG client.

Findings. Coverage focused on two policy levers: wage increases of up to P200 a day, and more social protection, including P43.18B released for cash aid and related programs. For the client that meant cost and wage pressure in the short term, a reputational risk around pricing, and, further out, room to line up with social protection priorities.

Tools: Excel, Media monitoring, Thematic coding, Data visualization
Code: github.com/liancataytayjk-coder/media-audience-engagement-analysis

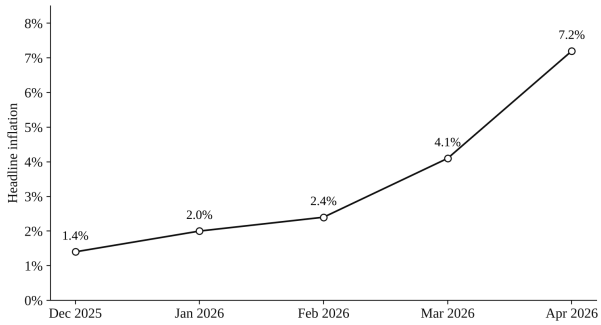


Figure 12. Philippine headline inflation, December 2025 to April 2026. *Source: Philippine Statistics Authority figures compiled in the media insight report.*